Summary in Graph

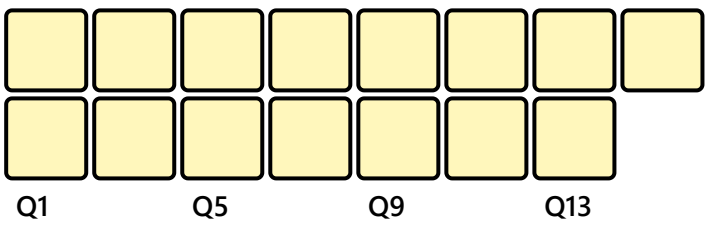
Exam Summary (GO Classes GATE CS | Computer Networks | Weekly Quiz 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (10 + 5)
Total Marks:	20 (10 + 10)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (45) out of 450 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Computer Networks

Consider the IP address $132.239.15.87$.

Suppose that we were still using class-based addressing and the network administrator had decided to break the network into 16 different subnets of equal size, What is the subnet address of the subnet to which this IP address would be attached?

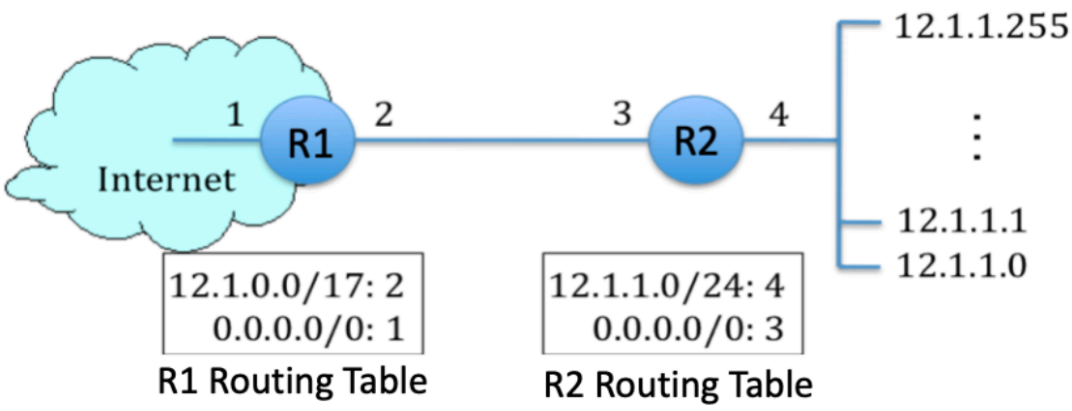
A. $255.255.240.0$

B. $132.239.0.0$
C. $132.239.15.0$
D. $132.239.240.0$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

A small university campus is assigned a large address block $12.1.0.0/17$, but is only using a portion of these addresses $(\text{in } 12.1.1.0/24)$ to number its computers.



The university network is connected to interface 4 of router $\mathrm{R2}$, Routing table of both routers are also given in the diagram.
Suppose the router $\mathrm{R1}$ receives a packet with destination IP address $12.1.20.1$. What path does this packet follow?

A. $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$
B. $1 \rightarrow 2 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow \ldots$
C. $1 \rightarrow 2 \rightarrow 3 \rightarrow 2 \rightarrow 1$
D. $1 \rightarrow 2 \rightarrow 1$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Numerical Type | Award: 1 | Penalty: 0 | Computer Networks

Imagine a world where IP addresses are 8 $(\text{not } 32)$ bits long. Suppose a router uses longest-prefix matching and has the following forwarding table:

Dst IP prefix	Output link
00	0
01	1
100	2
otherwise	3

What is the number of IP addresses reachable from Output link 3?

Your Answer: | Correct Answer: 96 | Not Attempted | Time taken: 00min 02sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Pretend for a moment that IP addresses are 12 bits instead of 32, and suppose you are allocated the block of addresses 1011xxxxxxx (i.e., all addresses that start with '1011' in binary notation). The network administrator has divided this network into 4 subnets as follows:

- Network A uses prefix $/5$
- Network B uses prefix /6
- Network C uses prefix $/7$
- Network D uses prefix $/8$

After the subnetting, which of the following is true?

A. There is only one unused block left of size 2^3
B. There is only one unused block left of size 2^4
C. There is only one unused block left of size 2^5
D. There is only one unused block left of size 2^6

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Numerical Type | Award: 1 | Penalty: 0 | Computer Networks

Consider the forwarding table of some router R as follows -

Prefix	Outgoing Interface
11010110 01100001 11111111 11000000	Port b
11010110 01100001 11111111 10000000	Port c
11010110 01100001 11111111 00000000	Port b
11010110 01100001 11111110 00000000	Port a

Suppose we perform route aggregation on the table (such that routing decisions are unchanged for incoming IP packets to R) then what will be the number of rows after aggregation?

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

What is the address range covered by the prefix $128.8.16.0/20$?

A. $128.8.16.0 - 128.8.16.255$
B. $128.8.16.0 - 128.8.20.255$
C. $128.8.16.0 - 128.8.24.255$
D. $128.8.16.0 - 128.8.31.255$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 1 | Penalty: 0 | Computer Networks

Consider the forwarding table of some router R as follows -

Prefix	Outgoing Interface
$128.2.0.0/24$	interface 1
$128.2.1.0/24$	interface 2
$128.2.2.0/24$	interface 2
$128.2.3.0/24$	interface 3

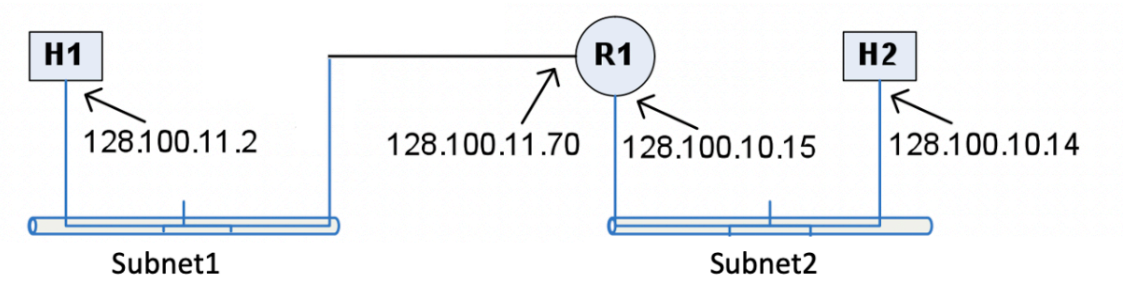
Suppose we perform route aggregation on the table (such that routing decisions are unchanged for incoming IP packets to R) then what will be the number of rows after aggregation?

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Consider two subnets connected to $\mathrm{R1}$ as shown below.

The interface IPs are also shown in the given diagram.



Please note that for every subnet all-zeros address is reserved for the network name, and the all-ones address is reserved for broadcast. The network administrator wants to set up subnet masks so that the network prefix has maximal length.

What should be maximal length subnet masks for both subnets?

A. $\mathrm{Subnet1}$: $128.100.11.0/25$ $\mathrm{Subnet2}$: $128.100.10.0/28$
B. $\mathrm{Subnet1}$: $128.100.11.0/25$ $\mathrm{Subnet2}$: $128.100.10.0/27$
C. $\mathrm{Subnet1}$: $128.100.11.0/26$ $\mathrm{Subnet2}$: $128.100.10.0/27$
D. $\mathrm{Subnet1}$: $128.100.11.0/24$ $\mathrm{Subnet2}$: $128.100.10.0/28$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Suppose a router has built up the routing table shown in the following table. The router can deliver packets directly over interfaces 0 and 1, or it can forward packets to routers $\mathrm{R2}$, $\mathrm{R3}$, or $\mathrm{R4}$. Assume the router does the longest prefix match.

Subnet Number	Subnet Mask	Next Hop
128.96.170.0	255.255.254.0	Interface 0
128.96.168.0	255.255.254.0	Interface 1
128.96.166.0	255.255.254.0	R2
128.96.164.0	255.255.252.0	R3
Default		R4

Which next hop should the router use for a packet destined to $128.96.167.151$?

A. Interface 0
B. Interface 1
C. R2
D. R3

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Computer Networks

Which of the following IP addresses belongs to the subnet $137.132.96/20$?

A. $137.132.96.96$

B. $137.132.104.104$
C. $137.132.112.112$
D. $137.132.120.120$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Computer Networks

An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it : $245.248.128.0/20$.
The ISP wants to give half of this chunk of addresses to Organization A, and a quarter to Organization B, while retaining the remaining with itself.

Which of the following is/are a valid allocation of addresses to A and B?

A. $245.248.128.0/21$ and $245.248.136.0/22$
B. $245.248.136.0/21$ and $245.248.128.0/22$
C. $245.248.128.0/21$ and $245.248.140.0/22$
D. $245.248.136.0/21$ and $245.248.132.0/22$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

You are given the following network prefixes:

$$171.64.121.0/24$$
$$171.64.180.0/24$$
$$171.64.18.0/24$$
$$171.64.181.0/24$$
$$171.64.19.0/24$$
$$171.64.187.0/24$$

You are to aggregate the prefixes together into the smallest possible number of shorter prefixes such that you do not cover any extra IPs.
What is the smallest number of prefixes, and what are they?

A. 1 prefix; $171.64.179.0/21$
B. 2 prefixes; $171.64.180.0/20, 171.64.18.0/21$
C. 4 prefixes; $171.64.180.0/23, 171.64.187.0/24, 171.64.121.0/24, 171.64.18.0/23$
D. 2 prefixes; $171.64.179.0/22, 171.64.183.0/24$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Computer Networks

Which of the following is/are True?

A. We can always do route aggregation irrespective of next hop is shared or not
B. If a routing table contains prefixes $31.75/16$ (for which packets are sent to port 1) and $31.75.93.128/25$ (for which packets are sent to port 2) then an arriving packet with IP address $31.75.93.129$ will be sent to port 2.

C. A routing table can correctly contain the two prefixes $128.50.128/17$ and $128.50.128/18$ simultaneously.
D. If a routing table is organized in order of increasing prefix length, then a routing decision may be performed by finding the last matching prefix.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Computer Networks

Consider three hosts with a given IP address and subnet mask.

IP Adress:	192.0.3.88	192.0.3.94	192.0.3.200
Subnet Mask:	255.255.255.0	255.255.255.240	255.255.255.240

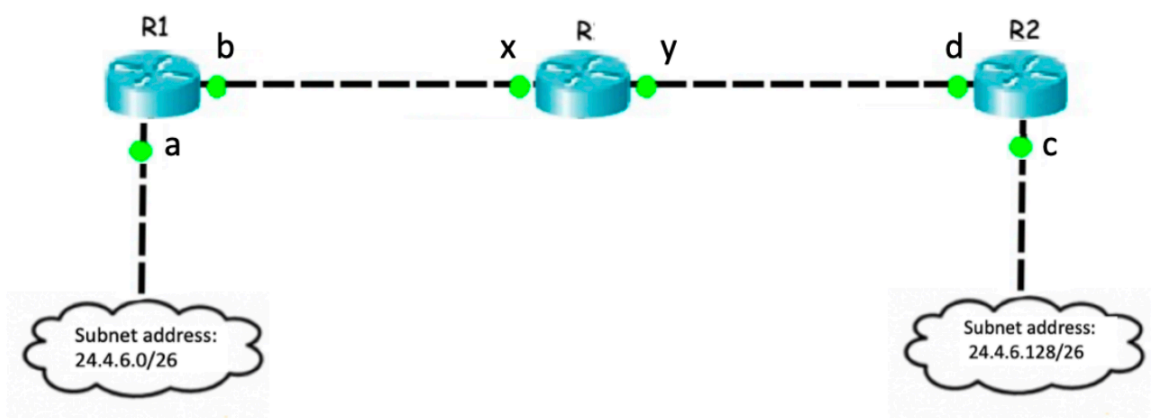
Which of the following is/are true?

A. Host C assumes that Host A is in the same network
B. Host A assumes that Host B is in the same network
C. Host B assumes that Host C is in the same network
D. Host B assumes that Host A is in the same network

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

Consider three routers $\text{R1}, \text{R2}, \text{and R3}$ connected as follows



As shown in the diagram, R1 is connected to the subnet with the subnet id as $24.4.6.0$, also R2 is connected to the subnet with the subnet id as $24.4.6.128$.

The netmask used in the network is $/26$.

What are the possible combinations of IP addresses of given interfaces a, b, x, y, c, and d?

Consider each option as an independent option.

A. $a : 24.4.6.1, b : 24.4.6.195, x : 24.4.6.250, y : 24.4.6.200$
B. $a : 24.4.6.1, b : 24.4.6.195, x : 24.4.6.200, y : 24.4.6.68$
C. $a : 24.4.6.5, x : 24.4.6.195, y : 24.4.6.250, d : 24.4.6.200$
D. $a : 24.4.6.1, b : 24.4.6.195, d : 24.4.6.250, y : 24.4.6.200$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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